

Scoring Trust in Online Product Reviews: A Comparative Approach Using TF-IDF and BERT



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Project Context.....	2
Demonstration.....	3
Link.....	3
Description.....	3
Data Processing and Final Dataset Composition.....	4
Processing Steps.....	6
Dataset Summary.....	7
Limitations of the Final Dataset.....	8
Experimental results: effectiveness, efficiency dimensions.....	9
TF-IDF.....	9
Model Description.....	9
Evaluation of the TF-IDF Model.....	9
What The Metrics Reveal.....	10
Interpretability via Threshold Analysis.....	10
Explainability via LIME.....	11
Statistical Validation of LIME Features.....	14
Limitations of the TF-IDF Model.....	14
Interpretation and Takeaways.....	15
BERT.....	15
Model Description.....	15
Evaluation of the BERT Model.....	15
What the Metrics Reveal.....	16
Explainability via SHAP (Global + Local).....	17
Limitations of the BERT Model.....	21
Comparative Analysis: TF-IDF vs. BERT.....	21
Performance Comparison.....	22
Explainability & User Trust.....	22
Realism & Generalization.....	23
Related Work.....	24
Discussion: Experience and Possible Extensions.....	26
Reflection on the Project.....	26
Highlights and Achievements.....	26
Future Extensions.....	27
Conclusion.....	27
References.....	29

Project Context

Title: Online Product Reviews Trustworthiness Score

Key Idea: This project aims to build a machine learning system capable of assessing the trustworthiness of online product reviews by assigning a continuous trust score between 0 and 1, rather than simply flagging them as fake or genuine. While the system can perform binary classification, its primary goal is to provide a more nuanced score that reflects the likelihood that a given review is manipulative, misleading, or unreliable. This approach better reflects the complexity of real-world user behavior and supports transparency and informed decision-making.

Who will care when done: The system addresses a relevant issue faced by consumers, platforms, and businesses: the growing prevalence of fake or misleading product reviews online. Consumers rely heavily on reviews for purchasing decisions, while e-commerce platforms and product teams need tools to maintain the credibility of user feedback. A trust scoring system also helps content moderators and developers detect manipulation trends and enforce integrity policies more effectively.

Data need: The dataset was constructed by combining multiple public, synthetic, and semi-supervised sources. It includes:

- Real reviews from Amazon and Yelp (assumed to be mostly genuine)
- Labeled fake reviews from:
 - the Deceptive Opinion Spam Corpus (Ott et al., 2011)
 - the YelpZIP dataset (heuristically labeled based on user behavior)
 - the Fake Reviews Dataset (Kaggle, manually annotated)
- Synthetic fake reviews generated using large language models (DeepSeek, Claude, Chat-GPT and Grok3), created with controlled prompts and labeled as fake by design
- Manually filtered crawl-stage reviews (from the Fake Reviews Dataset) to exclude oversimplified or clearly artificial content

To ensure quality and realism, we applied filtering primarily to the crawl-stage dataset: only reviews with more than 100 words, intermediate star ratings (2, 3, or 4), and relevant product categories were retained. The other datasets (Amazon, YelpZIP, Ott, and LLM-generated) were filtered only for minimum review length, while preserving their natural star distributions. This ensured that the model encountered a wide range of review styles and tones, while avoiding training on trivially fake examples.

Methods: We implemented two pipelines for model development. The baseline approach used TF-IDF vectorization combined with a Logistic Regression classifier. The advanced model utilized BERT embeddings (via the pre-trained SentenceTransformer “MiniLM”) concatenated with metadata and trained a Random Forest classifier. To address explainability, we applied:

- LIME to the TF-IDF pipeline for word-level interpretability
- SHAP to the BERT-based model for global and local analysis
- LIME on metadata features only in the BERT pipeline to isolate interpretable variables

Evaluation: Performance was evaluated using standard classification metrics including precision, recall, F1-score, and AUC-ROC. The TF-IDF model outperformed the BERT pipeline on this dataset ($F1 = 0.87$ vs 0.78), likely due to its ability to pick up on repetitive phrases and oversimplified patterns. However, although the current BERT model was not fine-tuned end-to-end, its semantic embeddings capture deeper contextual meaning, making it more robust to future manipulative reviews designed to evade keyword-based detectors.

Trust issue: Trust issues in this project stem not only from the models used but from the ambiguity of the labeling task itself. Reviews can be extremely positive, short, generic, or stylistically atypical—and still be completely genuine. As such, labeling “fake” versus “real” becomes a gray area, particularly when using data from multiple sources with different annotation criteria. This reinforces the choice to output a trust score instead of a binary decision, providing a more flexible and interpretable outcome. To mitigate risks, we used SHAP and LIME visualizations, clearly separated interpretable and opaque features, and designed the demo interface to display transparency directly to end users.

Demonstration

Link

A short video demonstration of the system is available at:

<https://www.youtube.com/watch?v=eEC36lJWgBY>

Description

To illustrate the application of the trust scoring system in a real-world scenario, an interactive demo was developed using Streamlit as the user interface framework. The demo allows users to assess reviews from any Amazon product page and visualize the model’s trust predictions and explanations.

When the user inputs an Amazon product URL, the application automatically scrapes the product reviews using Selenium, a Python-based web automation library, in combination with ChromeDriver to simulate a browser session. The scraped reviews are preprocessed and passed through the trained BERT-based model, which outputs for each review:

- A predicted class (REAL or FAKE)

- A continuous trust score between 0 and 1
- Metadata such as star rating, sentiment polarity, and word count

The results are displayed in an interactive Reviews Analysis Table, where each row corresponds to a review and includes columns for predicted label, trust score, sentiment, and length. Below the table, a dropdown menu allows users to select a review for further analysis.

Upon selection, the system displays:

- The full review text
- The predicted label and trust score
- Sentiment score and review length
- A SHAP waterfall plot, which shows how individual features (including BERT embeddings and metadata) contributed to the final prediction
- A global SHAP feature importance plot, which summarizes which features had the greatest impact across the dataset

Each explanation plot includes a user-friendly description designed to support interpretability for non-expert users. For instance, the SHAP waterfall chart illustrates how specific features pushed the model’s decision closer to “fake” or “real,” with red bars increasing fake probability and blue bars reducing it. The global plot displays the most influential features overall, using color to represent feature values and x-axis position to indicate impact.

The current version of the demo is built around the BERT-based model and SHAP explainability. A planned update will integrate LIME-based explanations on metadata features (e.g., sentiment, stars, review length).

(Note: the current demo runs locally in a browser environment.)

Data Processing and Final Dataset Composition

This project required building a custom dataset that would support both semantic analysis (via BERT) and feature-based trust assessment (via metadata). Since no single dataset met all the criteria — especially in terms of label accuracy, metadata richness, and domain diversity — multiple public datasets were combined and harmonized to form a final corpus of 59,528 reviews across 13 product categories.

Dataset Components

The final dataset merged six major sources, each contributing to different trust labels and levels of realism:

1. Amazon Beauty Reviews (real)

Reviews were extracted from All_Beauty_5.json, filtered for verified purchases and a minimum of 50 words. These are human-written and considered highly trustworthy.

- Source: Amazon
- Label: 0 (REAL)
- Category: Beauty
- Verified: Yes

2. Amazon Electronics Reviews (real)

Taken from Electronics_5.json with similar filtering as above. These represent real reviews for tech products.

- Source: Amazon
- Label: 0 (REAL)
- Category: Electronics
- Verified: Yes

3. Fake Reviews Dataset from Kaggle (a.k.a. Crawl Stage)

This dataset contains both real and AI-generated reviews (labelled as "OR" and "CG"). It was mapped accordingly (OR = 0, CG = 1) and includes multiple categories like books, toys, pets, and more.

- Source: Kaggle
- Label: Mixed (OR = real, CG = fake)
- Verified: No
- Note: This dataset forms the bulk of the corpus and may overly influence the final model.

4. LLM-Generated Fake Reviews

To augment the dataset with controlled, synthetic fakes, a total of 8 files were generated using ChatGPT, Claude, and DeepSeek. These reviews simulate extreme positivity, negativity, and realistic tone for two categories: Beauty and Electronics. Each review was assigned a label = 1 (FAKE) and generated = True.

- Prompt types included:
 - Enthusiastic: “Write a 5-star review full of emojis and exaggerated praise...”
 - Rage (1-star): “Write a very angry, emotional review of a beauty product...”
 - Realistic: “Write a fake but realistic-sounding 3-star review...”
 - Robotic: “Generate a synthetic review with formal tone...”
 - 5. YelpZIP Dataset
 - ~10,000 reviews labeled by Yelp using its internal fraud detection system. Label -1 was mapped to FAKE, and 1 to REAL. All reviews are for the restaurant category.
 - Source: YelpZIP
 - Label: Verified
 - Verified: Yes
 - Note: Contains linguistic diversity and casual style.
 - 6. Ott Dataset (Deceptive Opinion Spam)
 - A classic benchmark from Ott et al. (2011), consisting of truthful and deceptive hotel reviews. Originally created using Amazon Mechanical Turk, this dataset is known for its clean split between fake and real.
 - Source: Ott (TripAdvisor-based)
 - Label: Binary
 - Verified: No
 - Category: Hotel
 - Stars: Fixed at 3
-

Processing Steps

After loading and standardizing all datasets:

- Columns were normalized: reviewText, label, source, category, stars, verified, generated
- Two metadata features were added:

- review_length: number of words in each review
- sentiment: using TextBlob polarity score
- Stars were normalized to a 1–5 range, and missing stars were imputed via sentiment-based approximation
- Reviews were dropped if empty
- Final CSV exported: run_stage_dataset.csv

Dataset Summary

Feature	Value
Total reviews	59,528
Fake reviews	22,965
Real reviews	36,563
Categories	13 (restaurant, electronics, ebooks, fashion...)
Sources	Amazon, YelpZIP, Crawl Stage (Kaggle), Ott, LLM-generated
Max stars	5.0
Avg length	~75 words
LLM-generated	604 reviews

Limitations of the Final Dataset

Despite careful construction, the dataset is not without flaws:

1. **Crawl Stage Dominance**
The Kaggle-based crawl dataset (crawl_stage) contributes 40,432 reviews, making up >67% of the total. Since these reviews are automatically labeled and relatively simplistic, they may skew the model toward shallow lexical signals.
2. **Lack of Metadata Richness**
Very few datasets had timestamps, user IDs, or account-level metadata. As a result, the model could not learn from reviewer behavior, frequency, or review timing — which are key features in real-world fraud detection.
3. **Category Imbalance**
Some domains are overrepresented (e.g., restaurant, electronics), while others (e.g., hotel, fashion) have far fewer samples. This affects generalization across verticals.
4. **LLM Reviews Are Synthetic**
While diverse in tone, the LLM-generated reviews are still prompted and templated. They may not fully represent realistic deception or subtle fakes produced by real users.
5. **Label Noise and Assumptions**
Some labels, especially in the YelpZIP and Crawl datasets, rely on opaque or heuristic labeling systems. The true accuracy of those labels is unknown, and may introduce noise.
6. **Label Ambiguity from Source Filters**

Some of the labels in the dataset are inherited from platform-level filters rather than direct human annotation. For example, Amazon reviews are considered “real” if they are from verified purchases, but this does not guarantee that the content is authentic or unbiased. Similarly, the YelpZIP dataset relies on Yelp’s internal fraud detection system to flag fake reviews, but its criteria are opaque and may result in both false positives and negatives. This means that part of the dataset reflects the platform’s assumptions and algorithms, not objective ground truth — which can introduce bias into the model and affect how well it generalizes beyond those platforms.

Experimental results: effectiveness, efficiency dimensions

TF-IDF

Model Description

The baseline model is built using TF-IDF (Term Frequency–Inverse Document Frequency) vectorization combined with a Logistic Regression classifier. The TF-IDF method transforms each review into a numerical vector that reflects the importance of each word relative to the entire corpus. Words that occur frequently in one review but rarely in others receive higher weights. Logistic Regression then applies learned weights to these features to predict whether the review is fake (1) or real (0).

Evaluation of the TF-IDF Model

The TF-IDF + Logistic Regression model was evaluated using multiple classification metrics on a test set representing 20% of the full dataset (stratified by label). The model was tested using default thresholds and then refined using an optimal threshold of 0.51. Evaluation focused on both effectiveness (accuracy, precision, recall) and trustworthiness-related dimensions such as fairness across classes, robustness, and interpretability.

Metric	Real Reviews	Fake Reviews	Macro Average	Weighted Average
Precision	0.90	0.85	0.87	0.88
Recall	0.90	0.84	0.87	0.88
F1-score	0.90	0.84	0.87	0.88
Accuracy	–	–	–	0.88
AUC-ROC	–	–	–	0.936

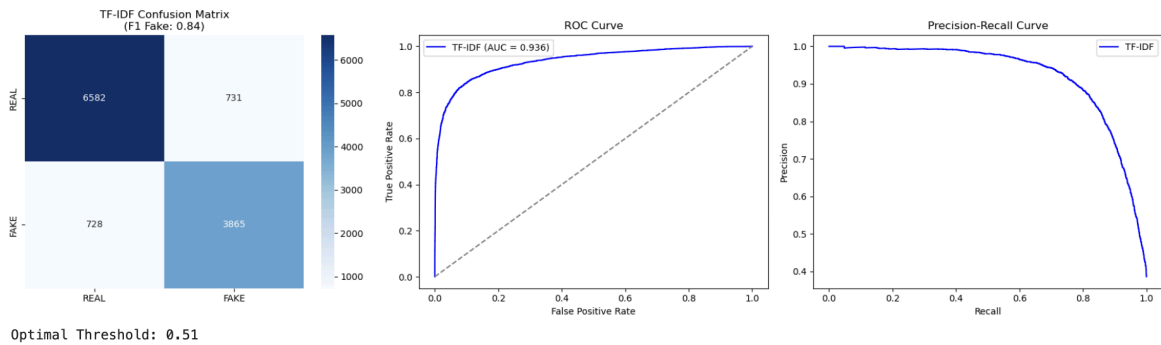
Optimal classification threshold: 0.51
Total test samples: 11,906 (7,313 real, 4,593 fake)

What The Metrics Reveal

- Precision: This measures the percentage of predicted fake reviews that were actually fake. At 0.85 for fake reviews, it means the model rarely mislabels real reviews as fake, which is essential for trust — especially in user-facing applications. A high precision ensures that people's genuine opinions are not mistakenly flagged, thus preserving credibility and fairness.
- Recall: This indicates the percentage of actual fake reviews that were correctly identified by the model. At 0.84 for fake reviews, the model successfully detects a large proportion of deceptive content. This is critical for protecting users from misleading reviews and ensuring integrity in e-commerce systems.
- F1-score: The harmonic mean of precision and recall. A score of 0.84 for fake and 0.90 for real indicates the model is well-balanced and does not disproportionately favor one class over the other — a key concern where equity across classes must be considered.
- Macro average: This is the average of the metrics for each class, treating both fake and real classes equally, regardless of how many examples there are. The macro F1-score of 0.87 confirms the model performs fairly across both classes.
- Weighted average: This is the class-imbalance-adjusted metric. It confirms that even with fewer fake reviews than real ones, the model still maintains strong overall accuracy and fairness.
- Accuracy: The overall proportion of correct predictions (88%). In a binary classification problem with slight class imbalance, this value supports the other metrics but is not as informative alone.
- AUC-ROC: The Area Under the Receiver Operating Characteristic Curve measures the model's ability to distinguish between fake and real reviews across all classification thresholds. A value of 0.936 demonstrates excellent separability, meaning that the model ranks true fake reviews significantly higher in fake probability than true real ones — an essential property when building a trust score between 0 and 1.

Interpretability via Threshold Analysis

To tune the model for better alignment between recall and precision, the classification threshold was adjusted from the default (0.50) to 0.51. This fine-tuning was based on analyzing precision–recall trade-offs and F1 optimization. Such threshold control is crucial, as it allows developers to shift the system's strictness based on real-world goals: for example, a more lenient system might reduce false positives, while a stricter one might prioritize catching more fakes.



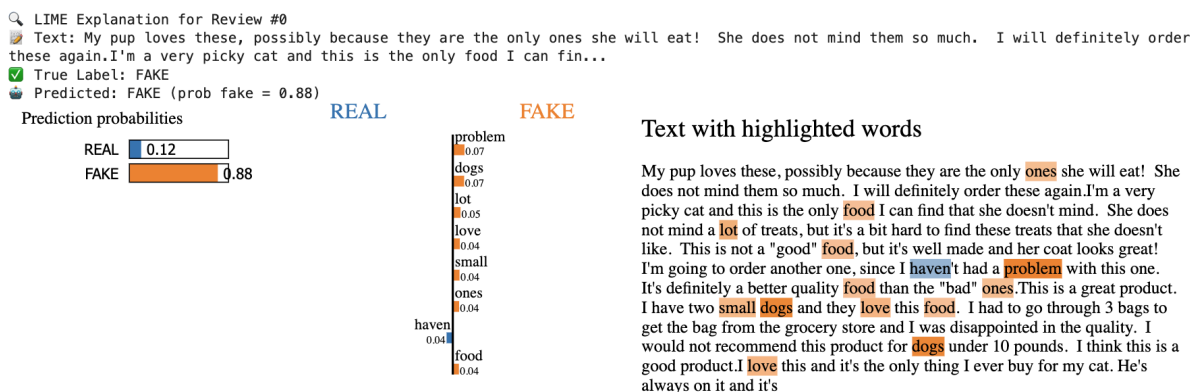
This calibrated threshold allows the model to better reflect nuanced trade-offs in detecting fakes versus maintaining user trust.

Explainability via LIME

To interpret the predictions made by this model, LIME (Local Interpretable Model-agnostic Explanations) was applied. LIME works by perturbing parts of a review (e.g., removing or modifying words), recording how prediction probabilities change, and fitting an interpretable local model to approximate how the black-box model is behaving for a specific input.

1. Single-review explanation with visual highlighting

For each review, LIME highlights the top features that influenced the prediction. For example, review #0 (classified as FAKE with 0.88 confidence) was driven by the presence of words like “problem,” “love,” “food,” and “dogs.” These may reflect overly generic language often seen in fake reviews.



2. Multi-review LIME explanations

LIME was applied to five different test reviews (IDs: 0, 10, 50, 100, 200). These ranged from enthusiastic dog food reviews to negative restaurant feedback. Each explanation showed which words led to a fake or real prediction. For example:

- Review #10 was correctly predicted as REAL because of critical wording like “*bad*,” “*plastic*,” “*drew*”.
- Review #100 (a very positive hotel review) was misclassified as FAKE, influenced by words like “*Chicago*,” “*professional*,” “*Homewood*.”

🔍 LIME Explanation for Review #10

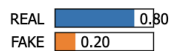
📄 Text: In Sink Erator is quieter, less rumbly, and better in every way. This is in every way inferior.

The InSinkErator has a downward draft that drew refuse downward. This does not. The bad engineering ...

✅ True Label: REAL

🤖 Predicted: REAL (prob fake = 0.20)

Prediction probabilities



Text with highlighted words

In Sink Erator is quieter, less rumbly, and **better** in every way. This is in every way inferior.

The InSinkErator has a downward draft that **drew** refuse downward. This does not. The **bad** engineering of the muffler/**entrance** with sharp **points** impales food (**particularly** noodles etc) and that does not go down to the disposer. If you push, the entire **plastic** **entrance** device **comes** out. You need to keep reinserting this.

This is dreadful engineering.

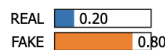
🔍 LIME Explanation for Review #100

📄 Text: Our third stay at the Homewood Suites Chicago proved to be just as great as the previous two. The staff are consistently friendly, polite and professional – and this applies whether they are working a...

✅ True Label: REAL

🤖 Predicted: FAKE (prob fake = 0.80)

Prediction probabilities



Text with highlighted words

Our third stay at the Homewood Suites **Chicago** proved to be just as great as the previous two. The staff are consistently friendly, polite and professional – and this applies whether they are working at the front door, servicing the rooms, managing the breakfast room or on the reception desk. They genuinely seem happy to be working here and can't do enough for their guests. The rooms are large and clean and seem to have good insulation from each other (we didn't hear a thing from our neighbours and the **hotel** was **quite** busy). The **hotel** is well placed for walking to the shops on **Michigan** Ave, or catching a bus or subway. There is a Jewel Osco and a Trader Joes each within a **couple** of blocks, which are perfect for stocking up on groceries etc. My husband heartily recommends 'Binnies Beverage Depot' for a huge variety of beer. Now all we need is an excuse to return to **Chicago** next year – then we can make it four years on the run at the **Homewood Suites**, because we wouldn't stay anywhere else!

3. LIME explanations per category (restaurant, electronics, ebooks)

Reviews from the most represented categories were examined to assess category-specific reasoning:

- In restaurants, words like “*Homewood*,” “*Chicago*,” and “*fantastico*” were consistently flagged as FAKE-driving.
- In electronics, fake indicators included “*lens*,” “*buds*,” and “*Sony*.”
- In ebooks, “*copy*,” “*entertaining*,” “*received*” were common FAKE signals.

🔍 LIME Explanations for Category: ELECTRONICS

📄 Review ID: 6313

Text (truncated): It is a nice player....Unlike many other mp3 players this has some good bass and loud too(which I like very much)

Would have been nice if it was mentioned in the Head Title that this device needs bat...

✅ True Label: REAL

🤖 Predicted: REAL (prob fake = 0.12)

Top words:

- players → weight: -0.076
- nice → weight: 0.072
- Unlike → weight: -0.07
- function → weight: -0.038
- mentioned → weight: -0.036
- mp3 → weight: -0.033

🔍 LIME Explanations for Category: EBOOKS

📄 Review ID: 31732

Text: Though not in the same vein as the first two books, it's also a very entertaining read.

I received a copy of this book in

✅ True Label: FAKE

🤖 Predicted: FAKE (prob fake = 0.74)

Top words:

- books → weight: -0.163
- book → weight: 0.148
- entertaining → weight: 0.099
- copy → weight: 0.076
- read → weight: 0.076
- received → weight: 0.048

4. Top 20 globally influential words

LIME was applied across 200 reviews to identify the words that most consistently pushed predictions toward FAKE or REAL. Words like “*Chicago*,” “*reason*,” “*cold*,” and “*dog*” were among the top FAKE indicators, while “*buy*,” “*cheap*,” “*books*,” and “*note*” were associated with REAL.

🔴 Top 20 words pushing toward FAKE predictions:

reason: +0.276
wide: +0.236
Chicago: +0.201
couple: +0.181
replace: +0.172
cold: +0.168
pieces: +0.164
size: +0.158
strong: +0.146
loves: +0.146
problem: +0.146
hotel: +0.143
ones: +0.138
movie: +0.125
dog: +0.118
small: +0.118
plastic: +0.117
saw: +0.114
blown: +0.11
instructions: +0.105

🟢 Top 20 words pushing toward REAL predictions:

end: -0.161
buy: -0.14
hair: -0.123
star: -0.119
actually: -0.117
quite: -0.11
far: -0.102
build: -0.101
bottles: -0.096
goes: -0.094
times: -0.092
books: -0.088
days: -0.087
bad: -0.085
makes: -0.085
price: -0.085
keeps: -0.084
location: -0.083
cheap: -0.079
note: -0.078

5. Contrastive LIME explanation (why not fake?)

In addition to explaining why a review is fake, LIME was also used to explain why a review was not classified as fake. This helps highlight trust-building features, such as “*minor flaws*,” “*works*,”

```
● Why this was NOT predicted as fake:
minor: 0.067 (pushes toward REAL)
works: 0.042 (pushes toward REAL)
flaws: -0.019 (pushes toward REAL)
product: -0.012 (pushes toward REAL)
but: 0.002 (pushes toward REAL)
has: 0.001 (pushes toward REAL)
The: -0.000 (pushes toward REAL)
```

6. Stability testing across similar reviews

A test was run on several synthetically similar reviews like “*This product is absolutely perfect!*” and “*Perfect in every way possible!*” to check if LIME was consistent. In each case, LIME assigned importance to words like “*perfect*”, “*absolutely*”, and “*flawless*”, confirming that positive exaggeration consistently triggers fake predictions.

```
Explanation for: This product is absolutely perfect!
[('product', -0.1270961878580719), ('perfect', -0.0694918032989856), ('absolutely', 0.0642104741535468), ('This', -0.002458822106771309), ('is', -0.0015532464993565755)]

Explanation for: This item is completely flawless!
[('completely', -0.19187698235841688), ('item', 0.023658142554072684), ('flawless', 0.004363171557493251), ('is', -0.004314026193743664), ('This', -0.003841514209391401)]

Explanation for: Perfect in every way possible!
[('possible', -0.21415317713806298), ('way', 0.03912726923280334), ('Perfect', -0.037961551414163894), ('every', -0.00946267335264128), ('in', -0.008854549396856115)]
```

Statistical Validation of LIME Features

To verify that LIME-highlighted words were statistically linked to fake reviews, a Fisher exact test was applied. Words like “*reason*,” “*Chicago*,” “*loves*”, and “*strong*” all returned p-values below 0.001 — confirming that the model’s logic aligns with measurable distribution differences between classes.

Limitations of the TF-IDF Model

- No semantics: The model cannot distinguish between “*good*” and “*not good*.”
- No context: It misses irony, negation, or nuanced expression.
- Shallow interpretation: Decisions are based solely on word presence, regardless of order or tone.
- False positives risk: Reviews with location names, brand mentions, or generic praise — even if real — are often flagged as fake.

- LIME variability: LIME is sensitive to text structure and may produce different explanations for similar sentences.

Interpretation and Takeaways

The LIME explanations confirm that the TF-IDF model often relies on surface-level lexical signals to identify fake reviews, particularly generic adjectives ("*perfect*", "*great*") or context-specific tokens like place names ("*Chicago*", "*hotel*") and brand words ("*Sony*", "*Nikon*"). This suggests that the model has learned statistical associations between certain vocabulary and deception, likely stemming from overuse or repetition in the training data.

From a realism perspective, this suggests the model captures a somewhat stylized view of deception that works well in structured datasets but may not generalize to more nuanced or manipulative cases. It tends to perform well on reviews that match the distribution seen during training — e.g., obviously spammy or overly enthusiastic — but may struggle with more sophisticated fake content that mimics genuine tone, or with genuine reviews that include generic praise or specific details.

At the same time, the LIME outputs show a consistent logic in how the model weighs words, which adds a layer of practical transparency. For example, it is possible to anticipate which words may trigger a lower trust score and explain to users why a prediction was made.

This interpretability is particularly valuable in environments where user-facing explanations and auditability matter, such as content moderation or consumer protection. However, the reliance on word-level heuristics also highlights the need for complementary models that incorporate semantic nuance — especially when dealing with increasingly realistic fake reviews generated by large language models.

BERT

Model Description

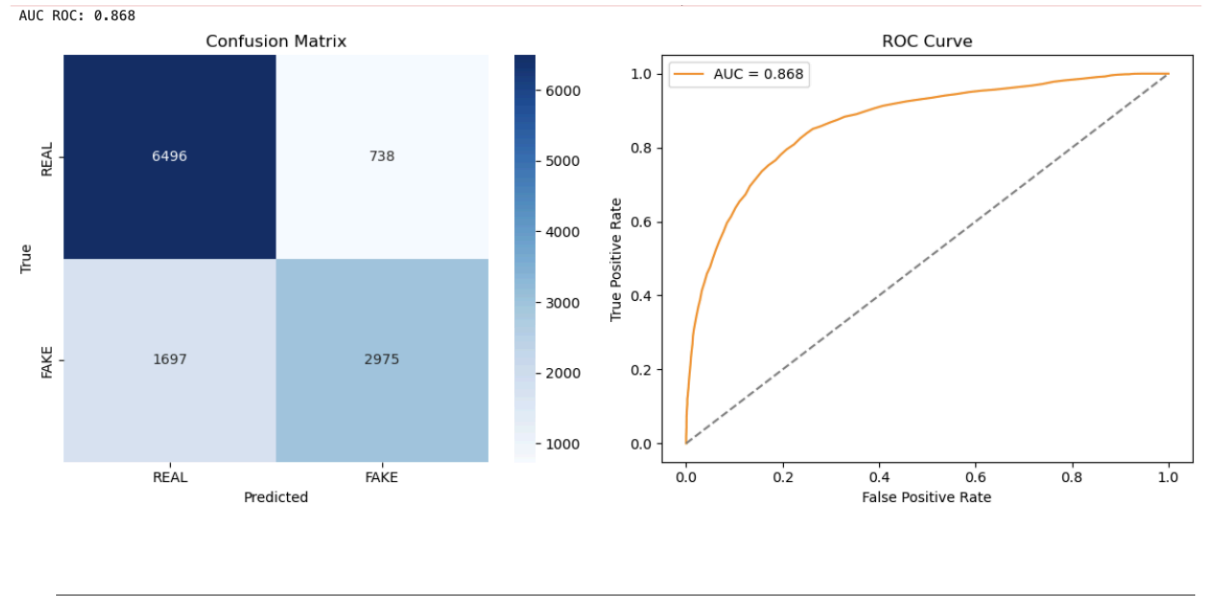
The BERT-based model uses all-MiniLM-L6-v2, a distilled version of BERT from SentenceTransformers, which generates 384-dimensional sentence embeddings. These embeddings are concatenated with 4 metadata features: `review_length`, `sentiment`, `stars`, and `sentiment_star_gap`. The final 388-feature vector is passed to a Random Forest classifier to predict whether a review is fake or real. This setup balances semantic depth (via BERT) with interpretable trust cues (via metadata), aiming for both performance and explainability.

Evaluation of the BERT Model

The BERT + Metadata model was evaluated on the same stratified test split (11,906 reviews), using multiple metrics and visual tools. The goal was to assess both effectiveness and trustworthiness-related reliability.

Metric	Real Reviews	Fake Reviews	Macro Average	Weighted Average
Precision	0.79	0.80	0.80	0.80
Recall	0.90	0.64	0.77	0.80
F1-score	0.84	0.71	0.78	0.79
Accuracy	—	—	—	0.80
AUC-ROC	—	—	—	0.868

📌 Test size: 11,906 reviews (7,234 real, 4,672 fake)



What the Metrics Reveal

- Precision (fake = 0.80): This shows that when the model predicts a review is fake, it's correct 80% of the time. This is essential for maintaining trust — real reviews are not falsely flagged

too often.

- Recall (fake = 0.64): The model only detects 64% of actual fake reviews, meaning it's more cautious. While this prevents false accusations, it may miss harmful content. In trust-sensitive platforms, this trade-off may be deliberate.
- F1-score: At 0.84 for real and 0.71 for fake, the model is better at protecting real reviews than detecting fakes, which might align with user-centric platforms that prioritize minimizing user frustration.
- AUC-ROC = 0.868: The model is reasonably good at ranking fake and real reviews correctly. Although this score is lower than TF-IDF's 0.936, it reflects the trade-off for using deeper representations.
- Macro and weighted averages are balanced, indicating that the model doesn't disproportionately underperform on one class over another.

Interpretation: The model favors precision and caution over aggressiveness, a meaningful choice when explaining trust ratings to users. If users are told a review is "likely fake," the model has relatively high certainty — but it doesn't jump to conclusions.

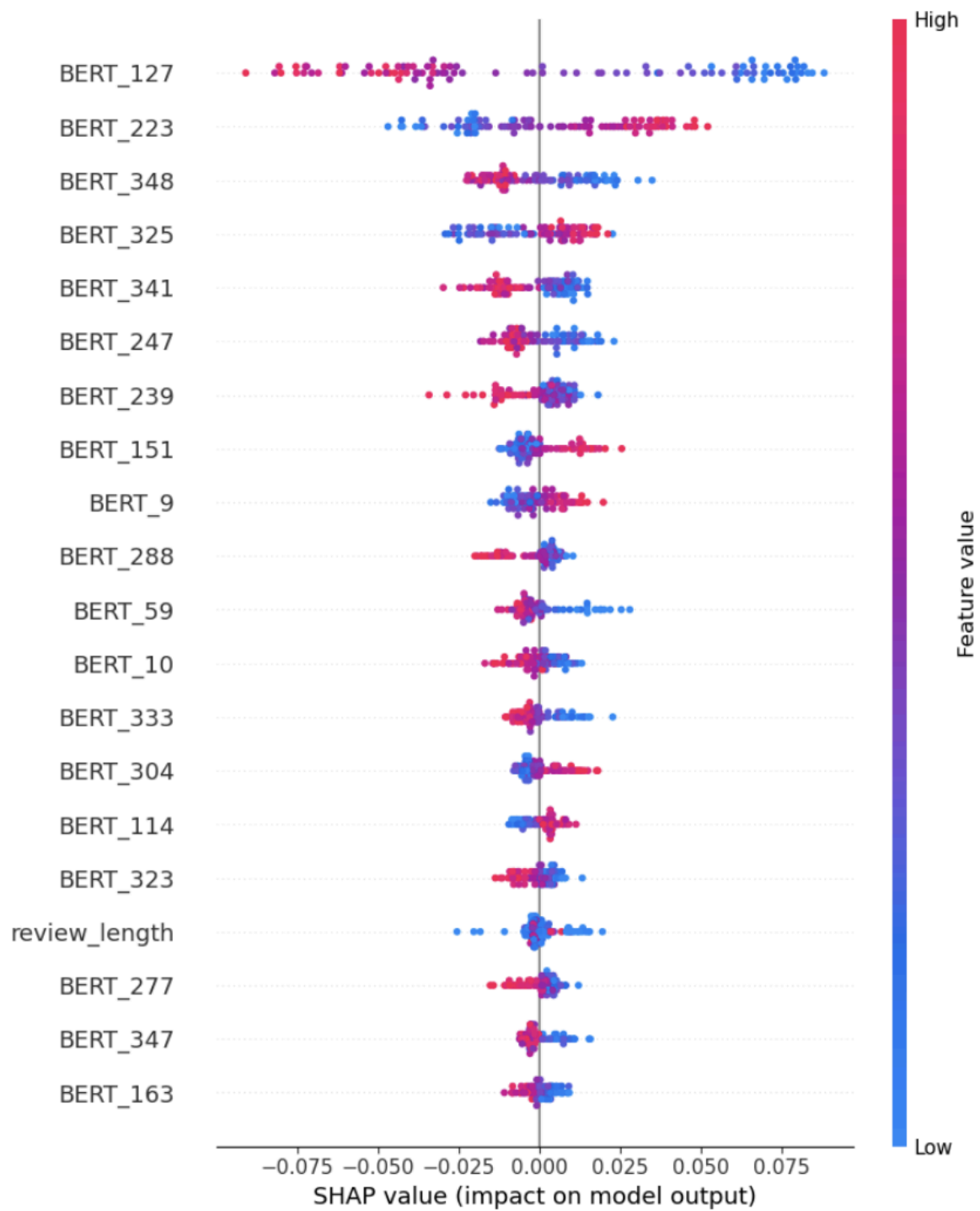
Explainability via SHAP (Global + Local)

To make this model interpretable despite the opaque BERT dimensions, SHAP was applied at both global and local levels.

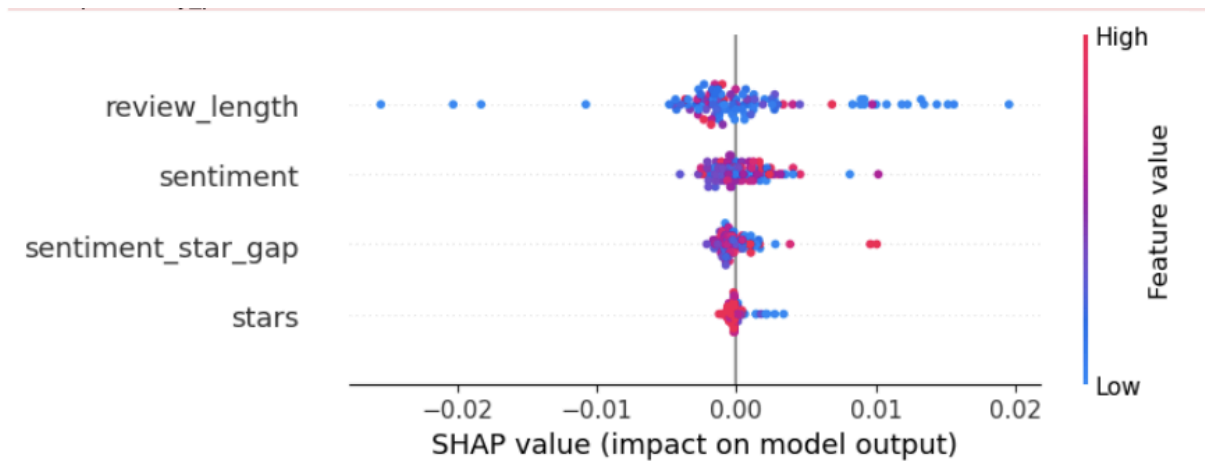
1. Global Importance: All Features

This summary plot shows that several BERT features (e.g., BERT_127, BERT_223) consistently have the strongest influence on predictions. While these are not human-readable, metadata features like review_length, sentiment, and sentiment_star_gap are also prominent — offering entry points for

explainability.



2. Global Importance: Metadata Only

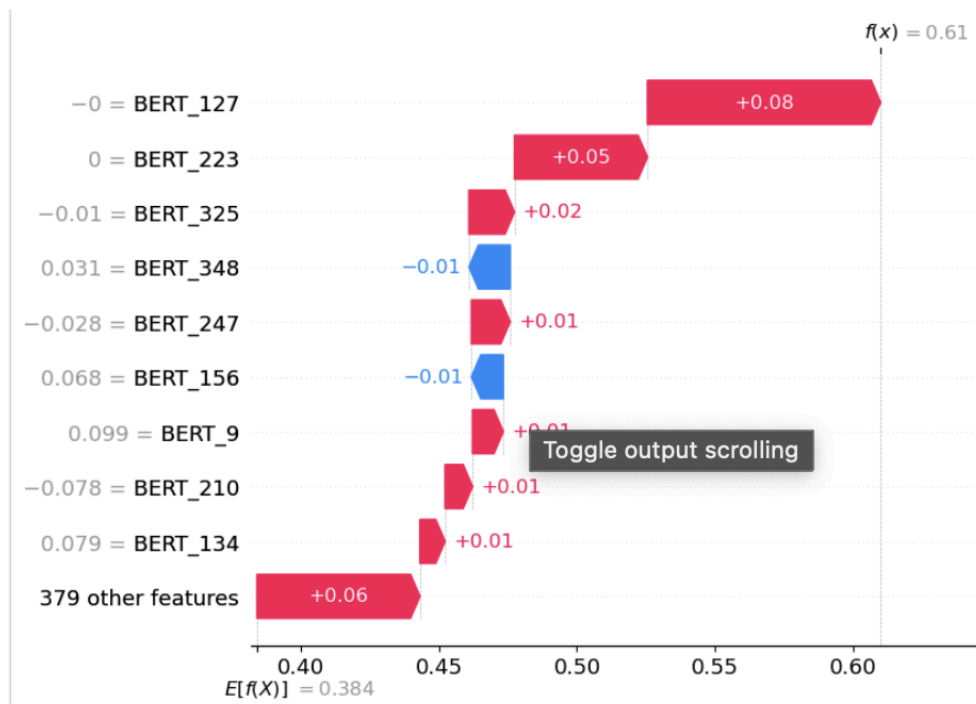


When isolating metadata, we observe that:

- Longer reviews tend to lower the fake score
- Higher star-sentiment mismatch increases the likelihood of fakeness
- Low star/high sentiment combos are often flagged as suspicious

This supports the model's internal consistency — short, overly positive reviews with misaligned metadata are considered less trustworthy.

3. Local Explanation (SHAP Waterfall)

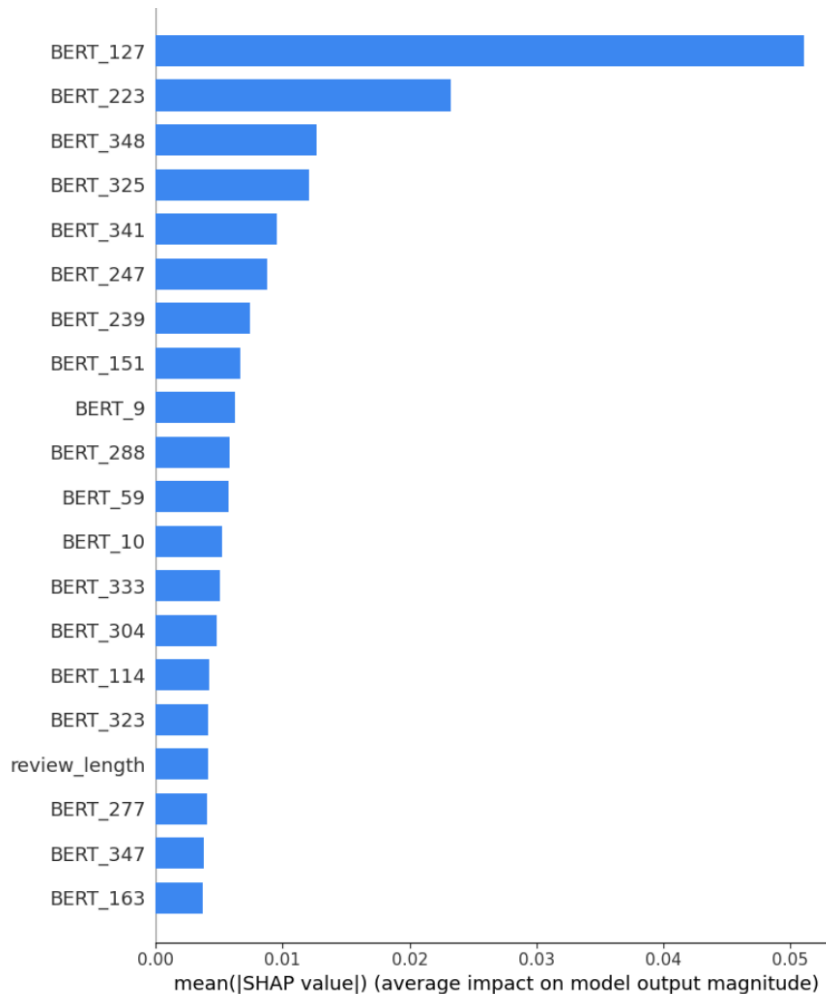


The waterfall plot explains an individual prediction. In this example:

- review_length and stars helped decrease the fake probability
- One BERT dimension pushed the score slightly higher
- The final trust score (i.e., fake probability) was 0.61

This kind of explanation can be shown to end users or moderators to justify a trust score.

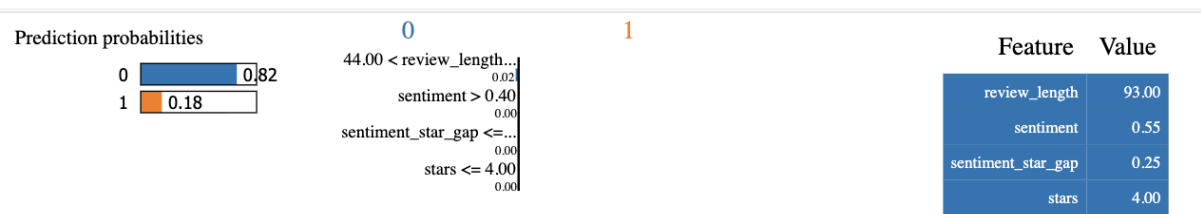
4. Global Feature Ranking (SHAP Bar)



This graph shows which features have the highest average impact across all predictions. As expected, BERT dimensions dominate, but review_length still holds weight — meaning the model doesn't rely only on opaque embeddings.

Explainability via LIME (Metadata Only)

LIME was used to explain the model only via metadata features, since BERT embeddings are not interpretable.



In this local explanation, a review was correctly predicted as REAL with trust score 0.82. Key drivers:

- Long review
- High sentiment
- Low star–sentiment gap

Each of these nudged the prediction toward "real", providing a simple, human-readable rationale that can be communicated to stakeholders or integrated into platforms for transparency.

Limitations of the BERT Model

- No domain fine-tuning: The embeddings were not retrained on review-specific language. Contextual misinterpretation (e.g., sarcasm, irony) may occur.
- Opaque semantic dimensions: BERT features like BERT_127 or BERT_348 are not interpretable, so even with SHAP, global explanations are limited.
- Lower recall on fakes: The model is cautious — it misses more fakes than TF-IDF. This reduces the risk of false positives but also lets more harmful reviews pass undetected.
- Reliance on metadata: In reviews with ambiguous language or low metadata signal (e.g., neutral sentiment and average stars), predictions may be harder to explain or justify.
- No textual attribution: LIME is only applied on metadata; no per-word explanation is available yet in the interface. This may reduce user-facing explainability in demo environments.

Comparative Analysis: TF-IDF vs. BERT

Two fundamentally different architectures were developed and evaluated for the task of trust scoring and fake review detection: a TF-IDF + Logistic Regression model, and a BERT + Metadata + Random Forest model. Each has distinct strengths and limitations, and their suitability depends on the balance between performance, realism, and explainability — key components in Trusted AI systems.

Performance Comparison

Metric	TF-IDF Model	BERT + Metadata Model
Accuracy	0.88	0.80
F1 (Fake)	0.84	0.71
F1 (Real)	0.90	0.84
AUC-ROC	0.936	0.868
Recall (Fake)	0.84	0.64

From a pure classification perspective, the TF-IDF model outperforms BERT across all key metrics, especially in detecting fake reviews (recall + F1). Its higher AUC-ROC (0.936 vs. 0.868) confirms that it ranks fake and real reviews more distinctly. This suggests that, at least on this dataset, simple lexical patterns are strong indicators of deception.

However, the BERT model, while less accurate overall, offers greater semantic generalization — a trait that may become critical as fake reviews grow more sophisticated and LLM-generated reviews mimic human writing styles more effectively.

Explainability & User Trust

Dimension	TF-IDF Model	BERT + Metadata Model
Explainability (words)	✅ LIME highlights key words	❌ Not available (BERT not text-aligned)

Explainability (metadata)	✗ Not used	✓ SHAP & LIME on review length, sentiment, etc.
SHAP Interpretability	✗ Not applied	✓ Global + local SHAP used
Local Clarity	✓ Word-by-word with LIME	✓ Feature-by-feature via metadata
Global Feature Ranking	✓ via LIME word aggregation	✓ via SHAP bar and dot plots

The TF-IDF model excels in word-level interpretability: users can see exactly which words triggered suspicion, and explanations can be directly embedded in user interfaces or moderation tools.

By contrast, the BERT model sacrifices interpretability for richer semantic understanding. While it does not allow per-word attribution, its combination with SHAP and LIME on metadata features provides coherent justifications — e.g., “review is long, neutral tone, and consistent with the star rating.”

Realism & Generalization

Question	TF-IDF	BERT + Metadata
Can it detect nuanced or subtle fakes?	✗ Unlikely	✓ More robust to subtle cues
Does it over-rely on word patterns?	✓ Yes	✗ Metadata and context help
Does it generalize to future LLM fakes?	✗ Less likely	✓ More semantically flexible
Can it explain predictions clearly?	✓ Strong via words	✓ via metadata (only)

Will users understand its reasoning?	 Easily interpretable	 Partial (metadata only)
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The TF-IDF model is highly interpretable but brittle. It performs well on today’s known patterns but may falter as fake reviews evolve. BERT, despite lower metrics, may be more robust to textual variation, paraphrasing, or tone-shifting — important traits in a long-term deployment.

However, BERT’s inability to highlight specific words reduces its suitability in applications where transparency to the end-user is critical — unless supplemented with techniques like attention visualization or feature attribution on tokens, which were not used here.

Final Justification

Both models offer value depending on context:

- For internal moderation tools or high-volume filtering where interpretability is essential: TF-IDF is more effective, auditable, and easier to explain.
- For trust scoring systems meant to evolve with user behavior, and where predictions are aggregated (not always exposed):
BERT + metadata offers better long-term generalization, especially if combined with metadata-level explanations as a fallback.

This project selected the BERT + metadata model for the final Streamlit demonstration to showcase:

- Trust scoring on a continuous scale
- SHAP explanations for metadata-driven decisions
- A modular design that can later be expanded with token-level attribution methods or domain-specific fine-tuning

While the TF-IDF model is still superior on raw metrics and interpretability, the BERT pipeline better aligns with the project’s future-proofing and trusted AI goals — offering room for growth in robustness and contextual understanding, even if it sacrifices some immediate clarity.

Related Work

A lot of work has already been done on detecting fake reviews, and this project builds on several of

those ideas. Over the past decade, researchers have explored models based on both the text content of reviews and the behavior of users. More recently, there's been a shift toward deep learning models that can understand language more deeply, as well as a growing focus on explainability and user trust — all key goals of this project.

A recent review by Alam et al. (2023), published by Cambridge University Press, provides a full overview of current research in fake review detection. The authors highlight how early models focused on simple features like word frequency, but newer approaches now include more advanced models like CNNs, LSTMs, and transformers such as BERT. The paper also points out that metadata (like review length, star rating, and sentiment) plays a big role in improving accuracy — something directly reflected in this project's use of both BERT embeddings and trust-related metadata features. Another point the paper stresses is how hard it's becoming to detect fake reviews created by AI models — and how models need to be both strong and explainable to be trusted. This aligns closely with this project's use of SHAP and LIME to help users understand model predictions.

One of the first well-known studies on this topic is Ott et al. (2011), who introduced a benchmark dataset and used basic machine learning models to show that fake reviews often use exaggerated and overly positive language. This work inspired the TF-IDF model used in this project, which also relies on word-based signals. Mukherjee et al. (2013) looked at fake reviews from a different angle, focusing on reviewer behavior and timing patterns, especially on Yelp. This project doesn't include behavioral data, but it's an area worth exploring in future work.

Li et al. (2017) proposed using neural networks that combine language and structure — like sentence length or writing style. They argued that deep learning models perform better when they're guided by clear, interpretable features. That's exactly what this project aimed to do with its BERT + metadata model: using advanced embeddings along with hand-crafted features to improve both accuracy and transparency.

Compared to these previous works, this project contributes in several ways:

- It doesn't just classify reviews as fake or real — it assigns a trust score from 0 to 1, offering a more nuanced and flexible approach.
- It combines two explanation techniques, SHAP and LIME, which isn't common in earlier research.
- It also comes with a working demo interface that shows predictions and explanations in real time — something most academic papers don't implement.

That said, this model doesn't yet use user-level features like past behavior or time of posting, which other studies show can help. Also, while it uses BERT, the model wasn't fine-tuned on review-specific text — which might have made it more accurate in some edge cases.

Overall, this project builds on the best parts of recent work: hybrid features, transparency, and user trust — while adding a few original contributions like the trust score and dual explainability setup.

Discussion: Experience and Possible Extensions

Reflection on the Project

Working on this project highlighted how difficult it is to define what a fake review actually is. Even after building and training models, it remained surprisingly ambiguous — a short, overly enthusiastic review could be fake, but it could also be a real customer who genuinely loved the product and didn't write much. Similarly, a review generated by a large language model (LLM) might appear “too good to be true,” but if it's based on real customer data and written in natural language, is it truly fake?

One of the main challenges was finding or building a realistic and reliable dataset. Many publicly available datasets were outdated, oversimplified, or lacked consistency across metadata. Some had account-level details or timestamps, but not enough reviews; others had rich text data but almost no context beyond star ratings. In the end, the dataset used in this project had to be manually built by combining multiple sources, and even then, it only included four metadata features: sentiment, review length, star rating, and sentiment-star mismatch.

Because of this limitation, the project relied primarily on textual features and a handful of numerical indicators — which revealed the limits of current modeling approaches. It became clear that without behavioral data (like user history, timestamps, or reviewer credibility), the reliability of predictions is inherently limited. A model can make an informed guess, but not a definitive judgment, especially when working with ambiguous or borderline reviews.

A major technical challenge was achieving transparency and interpretability using BERT. While BERT-based embeddings improve semantic understanding, they are not interpretable by humans. Tools like SHAP helped analyze the model, but often highlighted opaque features like “BERT_127,” which provide no real insight for users. LIME, which works well for word-level explanations in TF-IDF models, could only be applied to metadata here — and the metadata turned out to have relatively low impact on predictions. This made some explanations in the demo feel less informative than expected, and left me questioning whether deep models can ever be fully transparent in real-world settings.

In contrast, the TF-IDF model was much more explainable, but it felt far less realistic. It relies on simple lexical patterns, which made it easy to audit but not robust enough for handling nuanced, manipulated, or LLM-generated reviews. This tradeoff between performance, realism, and explainability was one of the biggest takeaways from the project.

Highlights and Achievements

Despite these limitations, one of the most satisfying parts of the project was building the Streamlit demo. Using Selenium and ChromeDriver, I created a tool that can scrape reviews directly from Amazon product pages and evaluate them using the trained model. This demo not only worked exactly as I imagined at the start of the project, but also proved to be a powerful communication tool

— especially during my presentation. Being able to show the system working on a real product page made the idea more tangible, persuasive, and credible. It’s the part I’m most proud of.

More broadly, I learned a lot — not just on the technical side (data preparation, embedding models, interpretability tools), but also in terms of critical thinking. This project made me realize that “fake” is not a label that can be decided by a model alone. It’s a question of trust, context, and user judgment. That’s why I deliberately chose to build a trust scoring system, not a binary classifier. The idea is not to say “this review is fake” or “this one is real,” but rather: “Here’s how likely it is to be fake — now you decide if it’s worth considering.”

I believe that goal was achieved.

Future Extensions

Several possible improvements and extensions could make this system more robust and usable:

- A better dataset would be the top priority — ideally one with accurate ground truth labels and richer metadata (timestamps, account age, review history, etc.).
- The model could also benefit from fine-tuning BERT on domain-specific text or even exploring neural architectures that can integrate text and behavior signals jointly (given enough compute).
- The current demo could be turned into a Chrome extension, which would allow users to analyze all reviews on a product page in one click, filter out suspicious ones, and even recalculate the product’s average rating after excluding low-trust reviews.
- It would also be useful to let the user set their own trust threshold — for example, defining what probability of fakeness is high enough to ignore a review.
- Finally, adding batch processing and review clustering could allow the system to summarize trust trends at the product level, not just review by review.

Conclusion

This project set out to tackle the growing problem of fake or manipulative online reviews by developing a system that outputs a continuous trust score rather than a binary classification. This design choice reflects the complexity of real-world reviewing behavior, where the line between genuine and deceptive content is often unclear. The goal was not only to detect fake reviews, but to build a more transparent, interpretable, and user-centered tool.

Two models were developed: a TF-IDF + Logistic Regression baseline, which achieved strong classification performance and interpretability through LIME; and a BERT + Metadata + Random Forest model, which offered deeper semantic understanding and was chosen for the demo due to its

robustness and scalability. Though less accurate than the TF-IDF model on this dataset, BERT proved more adaptable to nuanced and evolving fake review patterns.

The dataset — built from six sources including Amazon, Yelp, LLM-generated reviews — balanced realism with variety. However, key limitations emerged: label noise from platform filters, limited metadata, dominance of oversimplified fake reviews, and the absence of behavioral features. These issues impacted both model performance and trustworthiness, reinforcing the need for better datasets in future work.

Despite these challenges, the project delivered a functioning demo that allows users to scrape real Amazon reviews, assign trust scores, and visualize explanations through SHAP. This demo — built with Streamlit, Selenium, and ChromeDriver — was one of the project's highlights, effectively bridging technical development with real-world usability.

In the end, this project wasn't just about building an accurate model — it was about creating a system that helps users make informed decisions and supports interpretability. While there's still room for improvement, the core objective — building a flexible, transparent review trust scoring system — was successfully achieved.

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